
An Effective-Rank Audit of Alignment-Induced Activation Shifts: Confound Control, Constructive Calibration, and Limits

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Abstract

We audit alignment-induced shifts in residual-stream activations of three open-weight instruction-tuned LLMs (Llama-3.1-8B-Instruct, Gemma-2-9B-it, Qwen-2.5-7B-Instruct) using the effective rank of the alignment modification matrix on safety-relevant inputs, $\rho_\epsilon := \text{rank}_\epsilon(M_{\mathcal{D}_s})/d$, which formalizes the single-refusal-direction observation of Ardit et al. (2024) as a continuous quantity. The paper has three contributions. **(1) Confound-controlled measurement.** A four-variant decomposition of the modification matrix ($M_{\text{naive}}, M_{\text{template}}, M_{\text{aligned}}, M_{\text{DiD}}$) separates the contributions of chat-template formatting, alignment-stage shift, and the refusal-mediating direction, and recovers the Ardit refusal direction on M_{DiD} at $|\cos| \in \{0.77, 0.86, 0.50\}$ (Llama/Gemma/Qwen), while M_{template} gives chat-template-controlled $\rho_\epsilon \in \{0.0029, 0.0048, 0.0044\}$; the centered SVD residual is $4\text{--}7\times$ larger, indicating the headline rank is dominated by a single mean direction. **(2) Constructive calibration with a sweet-spot vs. brittle distinction.** On a 3-layer MLP across $\rho_\epsilon \in \{0.008, 0.17, 0.33, 0.40\}$, high ρ_ϵ via mild rank-maximization regularization ($\lambda = 5$) buys ablation robustness (≥ 0.97 compliance through projection of 62.5% of the representation), while strong regularization at the same nominal ρ_ϵ ($\lambda = 50$) does not (0.10 at $r = 40$). ρ_ϵ is therefore a diagnostic for fragility, not a target whose mechanical inflation buys robustness. Linear-projection fragility (Prop. 3.1, unconditional) and an $O(\rho_\epsilon d)$ achievability bound on faking footprint (Prop. 3.5, representational only) are the two structural consequences of low ρ_ϵ . **(3) Limits of rank-based diagnostics.** (a) The diagnostic is not safety-specific (LRH baseline $\rho_\epsilon \approx 0.008\text{--}0.010$, only $2\text{--}3\times$ the safety value); (b) SVD principal ordering does not match causal ordering (Llama narrow-band u_2 ranks second by σ but is causally inert; cumulative ablation is non-monotone at $k=5$); (c) the spectral-gap hypothesis required to upgrade Prop. 3.5’s achievability to a matching $\Omega(\rho_\epsilon d)$ Mirsky-route lower bound fails empirically ($\kappa_{\text{emp}} > 8$ on 1/90 Llama layer-reference pairs and 0/36 MLP combinations) and structurally (Lemma 8.1: $\kappa_{\text{lb}} \leq 2/(\epsilon r)$, so the hypothesis $\kappa > 8$ fails once $\epsilon r > 1/4$). The matching lower bound remains an open problem (§8).

1 Introduction

Arditi et al. (2024) showed refusal in instruction-tuned LLMs is mediated by a single direction in activation space, and that ablating that direction nearly eliminates refusal of harmful instructions. This is a striking empirical observation about the geometry of current alignment: the shift between pre-trained and aligned activations on safety-relevant inputs lives in a small number of directions, and a sub-capable adversary with a truncated SVD can identify and remove them.

This paper audits the structural consequences of that observation by formalizing it as a continuous quantity. Let $M_{\mathcal{D}_s} := [h_{\text{align}}(x_i) - h_{\text{pre}}(x_i)]_i$ be the paired alignment modification at a fixed layer over a safety-relevant input distribution $\mathcal{D}_s$. The *separability index* is

$$\rho_\epsilon := \text{rank}_\epsilon(M_{\mathcal{D}_s})/d$$

(§2), the effective rank of $M_{\mathcal{D}_s}$ normalized by the representation dimension. On Llama-3.1-8B-Instruct, Gemma-2-9B-it, and Qwen-2.5-7B-Instruct the chat-template-controlled ρ_ϵ is 0.0029, 0.0048, 0.0044; the centered SVD residual is 4–7 $\times$ larger than the uncentered, so the index is dominated by a single mean direction that aligns with the Arditi refusal direction at high cosine after control-distribution subtraction. The paper makes three contributions.

Contribution 1: confound-controlled measurement. A four-variant decomposition of the modification matrix ($M_{\text{naive}}, M_{\text{template}}, M_{\text{aligned}}, M_{\text{DiD}}$; §6.1) separates the contributions of chat-template formatting, the alignment-stage shift, and the refusal-mediating direction. The chat-template-controlled M_{template} is the cleanest quantity for the separability index; M_{DiD} subtracts the control column-mean and recovers Arditi’s intra-aligned refusal direction at $|\cos| \in \{0.77, 0.86, 0.50\}$ (Llama, Gemma, Qwen), high in Llama and Gemma, moderate in Qwen—the Arditi rank-1 collapse is not universal but holds on two of three families. M_{template} and M_{DiD} measure structurally different quantities (alignment-stage shift vs. refusal-mediating direction); the four-variant view makes the distinction explicit and locates the chat-template confound that inflates uncontrolled ρ_ϵ estimates by $\approx 4\times$ (App. A.1).

Contribution 2: constructive calibration with a sweet-spot vs. brittle distinction. On a 3-layer MLP across $\rho_\epsilon \in \{0.008, 0.17, 0.33, 0.40\}$ (§5, App. A.9), high ρ_ϵ via mild rank-maximization regularization ($\lambda = 5$, baseline compliance 0.86) buys ablation robustness (≥ 0.97 compliance through projection of 62.5% of the representation), while strong regularization at the same nominal ρ_ϵ ($\lambda = 50$, baseline compliance 0.71) does not (post-ablation compliance 0.10 at $r = 40$). ρ_ϵ is therefore a *diagnostic*; mechanical rank inflation via stronger regularization does not buy structural robustness, so treating ρ_ϵ as an optimization target fails on the calibration testbed. Two structural consequences of low ρ_ϵ support the diagnostic interpretation: linear-projection fragility (Prop. 3.1, unconditional) and an $O(\rho_\epsilon d)$ achievability bound on faking footprint (Prop. 3.5, representational only). The matching $\Omega(\rho_\epsilon d)$ lower bound, which would make low ρ_ϵ a structural *condition* for fake-resistant safety, is left as an open problem (§8).

Contribution 3: limits of rank-based diagnostics. We document three. **(a) Not safety-specific.** The LRH baseline (App. A.7) gives arbitrary linear concepts $\rho_\epsilon \approx 0.008\text{--}0.010$ in the same models, only 2–3 $\times$ the safety value: low rank is a generic property of post-training linear concepts, and the bridge from rank to behavior is supplied by causal ablation (Test 2, §6.4), not by the rank number. **(b) SVD ordering does not match causal ordering.** On Llama narrow-band, u_2 ranks second by squared variance but has near-zero causal effect on refusal under projection ablation; cumulative top- k ablation is non-monotone at $k = 5$ (Table 6, §6.4). Behavioral fragility must be measured directly by Test 2, not read off the SVD spectrum. **(c) The spectral-gap hypothesis required for the Mirsky lower-bound sketch fails.** We measure κ_{emp} across three publicly available unsafe-mode references for Llama (§8, App. A.10): $\kappa_{\text{emp}} > 8$ on 1/90 layer-reference pairs. In a controlled MLP sweep (App. A.11), 0/36 honest–unsafe combinations cross the threshold, with the worst case in the distributed-vs-distributed regime the high- ρ_ϵ regime targets. We identify a geometric reason (Lemma 8.1: $\kappa_{\text{lb}} \leq 2/(\epsilon r)$) that $\kappa > 8$ is structurally hard to satisfy whenever the modal-difference signal has non-trivial effective rank.

The diagnostic is computable by a verifier strictly weaker than the diagnosed system, given whitebox paired checkpoints and a fixed $\mathcal{D}_s$ (Remark 3.4); the capability burden is displaced onto constructing $\mathcal{D}_s$, not removed.

2 The Separability Index

Let $f_{\text{pre}}, f_{\text{align}}$ be the pre- and post-alignment systems, $h_{\text{pre}}(x), h_{\text{align}}(x) \in \mathbb{R}^d$ their layer activations, and $\mathcal{D}_s, \mathcal{D}_c$ distributions of safety-relevant and control inputs. The alignment modification matrix on a sample $x_1, \dots, x_n \sim \mathcal{D}$ is

$$M_{\mathcal{D}} := [h_{\text{align}}(x_1) - h_{\text{pre}}(x_1), \dots, h_{\text{align}}(x_n) - h_{\text{pre}}(x_n)] \in \mathbb{R}^{d \times n}. \quad (1)$$

With singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$ of $M_{\mathcal{D}}$, the *effective rank* at tolerance $\epsilon_{\text{rank}} \in (0, 1)$ is

$$\text{rank}_{\epsilon_{\text{rank}}}(M_{\mathcal{D}}) := \min \{ k : \sum_{i=1}^k \sigma_i^2 \geq (1 - \epsilon_{\text{rank}}) \sum_i \sigma_i^2 \}. \quad (2)$$

We write ϵ when unambiguous and reserve ϵ_{pert} for the perturbation-norm tolerance in §8. We use $\epsilon_{\text{rank}} = 0.05$. Since $M_{\mathcal{D}} \in \mathbb{R}^{d \times n}$ has $\text{rank} \leq \min(d, n)$, when $n < d$ the sample-relative ratio $\text{rank}_{\epsilon} / \min(d, n)$ is the more conservative summary; §6 reports both and sweeps n .

The question is not whether f_{align} produces safer outputs (assumed) but how the modification is distributed in representation space.

2.1 Separable Safety

Definition 2.1 (Separable Safety). An alignment procedure produces (k, ϵ) -separable safety on safety-relevant distribution $\mathcal{D}_s$ if the alignment modification is concentrated in a low-dimensional subspace:

$$\text{rank}_{\epsilon}(M_{\mathcal{D}_s}) \leq k, \quad k \ll d. \quad (3)$$

Equivalently, the top- k principal components of $M_{\mathcal{D}_s}$ capture at least a $(1 - \epsilon)$ -fraction of the modification’s variance. Safety is *perfectly separable* when the alignment shift is rank-one ($\text{rank}_{\epsilon}(M_{\mathcal{D}_s}) = 1$ for small ϵ).

Separability is a statement about the *readout* of the alignment modification at the residual stream, consistent with arbitrarily complex internal computation producing the readout (Remark 2.2).

Remark 2.2 (Modification Level vs Mechanism). $M_{\mathcal{D}_s}$ records what alignment changes at the readout, not the internal computation that produced the change. Mechanistic interpretability (Olah et al., 2020; Ardit et al., 2024; Wollschläger et al., 2025) consistently finds distributed non-linear computations writing into low-dimensional residual subspaces; our diagnostic is sensitive to that readout structure and silent on the depth of computation. Fragility, fakeability, and verifiability claims thus concern residual-stream attacks (projection, steering, mode-toggling), not arbitrary adversarial pressure on the mechanism.

Remark 2.3 (Linear Recoverability). Low rank means a rank- k projection P recovers $(1 - \epsilon)$ -energy on P and removes it on $I - P$; a linear probe with $\leq k$ features achieves near-perfect aligned-vs-pre-trained classification when the shift exceeds activation noise. Non-linear probes may find additional structure outside SVD.

Low ρ_{ϵ} has one unconditional geometric consequence (Prop. 3.1: a rank- k projection removes $(1 - \epsilon)$ -variance) and one unconditional representational consequence (Prop. 3.5: an $O(\rho_{\epsilon}d)$ faking footprint is achievable). Whether the representational fragility translates to behavioral fragility, and whether the achievability has a matching Ω lower bound, are separate empirical (§6.4) and open-problem (§8) questions.

2.2 The Separability Spectrum

The *separability index* is

$$\rho_{\epsilon}(M_{\mathcal{D}_s}) := \text{rank}_{\epsilon}(M_{\mathcal{D}_s})/d \in (0, 1]. \quad (4)$$

At $\rho_{\epsilon} \ll 1$ the modification inherits fragility and low-cost-faking (§3); at $\rho_{\epsilon} \rightarrow 1$ alignment spans the full representation (constructing such an LLM alignment is open).

ρ_{ϵ} is structural, not behavioral: two systems with identical $\mathcal{D}_s$ compliance can have very different ρ_{ϵ} (§5). Its diagnostic content is projection-ablation response, not nominal behavior.

3 Consequences of Separability

The diagnostic carries one geometric implication unconditionally; behavioral and converse-direction implications require additional structure that this paper shows is not free (§8).

3.1 Fragility

Proposition 3.1 (Linear Fragility). *If alignment is (k, ϵ) -separable on $\mathcal{D}_s$, an adversary with whitebox activation access and the pre-trained baseline can, given a finite sample sufficient to estimate the top- k principal subspace (empirically $n=200$ suffices on $d \geq 3584$; see App. A.2 for bootstrap stability checks and App. A.4 for sample-size sensitivity), remove $(1 - \epsilon)$ -variance by a rank- k projection.*

Proof. $M_{\mathcal{D}_s} = U\Sigma V^\top$, U_k the top- k left singular vectors, $P_k := U_k U_k^\top$. (k, ϵ) -separability gives $\sum_{i \leq k} \sigma_i^2 \geq (1 - \epsilon) \sum_i \sigma_i^2$. Replace $h_{\text{align}}(x)$ with $(I - P_k)h_{\text{align}}(x) + P_k h_{\text{pre}}(x)$. U_k from a truncated SVD on a sample. $\square$

Remark 3.2 (Threat Model). Substitution requires open-weight access to h_{pre} . The weaker form $h_{\text{align}} \rightarrow (I - P_k)h_{\text{align}}$ needs no such access but only shows the principal subspace is causally privileged; §6 uses the weaker form. LoRA fine-tuning and prompt jailbreaks are out of scope.

Geometric vs. behavioral fragility. SVD orders by variance, not causal authority: in Llama narrow band u_2 has near-zero causal effect despite ranking second by σ (Table 6). Test 2 (§6.4) is the bridge: in Gemma narrow $k = 3$ the principal and causal subspaces coincide; in Llama they don’t. We do not state a behavioral-fragility proposition—no Lipschitz hypothesis holds on the principal parametrization, and a causal-subspace hypothesis is empirically circular.

Remark 3.3 ($k = 1$ and Arditi). At $\text{rank}_\epsilon \approx 1$, u_1 coincides with $\bar{\Delta} = \mathbb{E}[h_{\text{align}} - h_{\text{pre}}]$. Arditi et al. (2024)’s direction is the distinct within-aligned quantity $\bar{\Delta}^{\text{Arditi}} = \mathbb{E}_{\mathcal{D}_s}[h_{\text{align}}] - \mathbb{E}_{\mathcal{D}_c}[h_{\text{align}}]$. Cosines in §6.

Remark 3.4 (Computability by a sub-capable verifier). ρ_ϵ is computable by truncated SVD on paired-checkpoint activations alone. The capability burden is displaced onto $\mathcal{D}_s$, not removed: a super-capable system can produce safety-relevant inputs humans cannot recognize, and a system optimized on $\mathcal{D}_s$ may look safe outside it.

3.2 Cost of Faking: Achievability

Fragility (§3.1) says low ρ_ϵ *permits* an adversary to project out the alignment shift. A symmetric question about *faking* alignment—can a deceptive policy exhibit honest activations under monitoring while behaving differently when unmonitored—has the same representational-footprint upper bound, also unconditional.

Proposition 3.5 (Representational Achievability). *Let $r := \text{rank}_\epsilon(M_{\mathcal{D}_s}^{\text{honest}})$ and P_r the projector onto its top- r principal subspace. The mode-toggling policy $h^{(m)}(x) = h_{\text{pre}}(x) + m \cdot P_r(h_{\text{align}}(x) - h_{\text{pre}}(x))$, $m \in \{0, 1\}$, matches the projected honest shift $P_r M_{\mathcal{D}_s}^{\text{honest}}$ in monitored mode and has rank- r modal difference: representational footprint $O(\rho_\epsilon d)$. (Proof in App. A.13.)*

The statement is unconditional but weak: it says low ρ_ϵ *permits* a representationally-cheap shift matching the projected honest activations on $\mathcal{D}_s$, not a policy that passes any non-trivial behavioral monitor. The matching $\Omega(\rho_\epsilon d)$ lower bound—which would yield the contrapositive “high ρ_ϵ blocks low-cost mode-toggling”—is left as an open problem (§8); a natural Mirsky-style sketch requires a spectral-gap hypothesis we show fails both empirically and structurally.

4 The Three Diagnostic Tests

Three tests on $\mathcal{D}_s$ at a chosen layer band: **Test 1 (rank)**: SVD on $M_{\mathcal{D}_s}$, report rank_ϵ at $\epsilon = 0.05$ ($\mathcal{D}_c$ as control). **Test 2 (causal ablation)**: project out $P_k = U_k U_k^\top$, measure refusal change vs. a random rank- k null. **Test 3 (principal alignment)**: $|\cos(u_1, \bar{\Delta})|$ on $\mathcal{D}_s$.

Low ρ_ϵ predicts (T1) steeply rising cumulative variance, (T2) sharp refusal collapse on principal-but-not-random ablation, (T3) high cosine. Test 2 is the empirical bridge: SVD ordering does not match causal ordering globally (§6.4), so projection effect must be measured.

5 Calibrating the Diagnostic

We calibrate on a 3-layer MLP ($d_{\text{hidden}} = 128$, input dimension $d_{\text{in}} = 20$, $N_{\text{train}} = 4000$). The task is a binary classification $y_{\text{task}} = \mathbf{1}[X_4 X_5 - X_6 X_7 > 0]$ on $X \sim \mathcal{N}(0, I_{20})$; the safety distribution $\mathcal{D}_s$ is the disjoint corner $\{X_1, X_2, X_3 > 1\}$ (30% of training data by design), on which the safe label is forced to $y_{\text{safe}} = 0$. The safety-defining features (X_1, X_2, X_3) and the task-defining features (X_4, X_5, X_6, X_7) do not overlap. Compliance on $\mathcal{D}_s$ is the fraction of probe inputs in $\mathcal{D}_s$ where the network outputs 0. We train three alignment procedures: **Steering** (rank-1 activation overlay gated to $\mathcal{D}_s$); **Full-fine-tune** (FT on y_{safe}); **Distributed** (FT + rank-max regularizer: stable-rank surrogate $\|M\|_F^2 / \|M\|_2^2$ plus column-orthogonality, App. A.8). Stable rank is a

differentiable surrogate for rank_ϵ at training; the two coincide in flat-spectrum limit and diverge for power-law. Aggregate ρ_ϵ on $\mathcal{D}_s$ (mean over 5 seeds) is $\{0.008, 0.173, 0.401\}$ across variants. The Steering and FullFT variants have baseline compliance ≥ 0.93 ; Distributed achieves the highest ρ_ϵ but at a compliance cost (baseline 0.71–0.94 depending on λ). We report the rank-vs-compliance trade-off as a structural fact rather than normalizing it away: the three variants are not at matched compliance, and the diagnostic’s calibration content lives in how ρ_ϵ moves across the procedures, not in treating them as exchangeable.

Finding: rank inflation does not buy robustness. Within the Distributed family, sweeping $\lambda \in \{0.5, 5, 15, 50\}$ (App. A.9, Table 1) reveals that the same nominal $\rho_\epsilon \approx 0.40$ admits two qualitatively different alignments. At $\lambda = 5$ the alignment is ablation-robust: compliance remains at 1.00 even after projecting out 62.5% of the representation ($r = 80$, $d = 128$). At $\lambda = 50$ the same nominal ρ_ϵ produces an ablation-brittle alignment: compliance collapses to 0.10 at $r = 40$. Baseline compliance also drops monotonically with λ ($0.94 \rightarrow 0.86 \rightarrow 0.78 \rightarrow 0.71$), so mechanical rank inflation buys neither robustness nor compliance once the regularizer is strong enough.

Table 1: Same ρ_ϵ , opposite ablation response in the Distributed family. Compliance after projecting top- r principal directions of $M_{\mathcal{D}_s}$; MLP $d = 128$, mean $\pm$ std across 3 seeds; baseline is pre-ablation compliance on $\mathcal{D}_s$. Source: `calibration_v2_revision3.json`.

λ	ρ_ϵ	Baseline	Compliance at $r = 40$	Compliance at $r = 80$
0.5	0.33 $\pm$ 0.00	0.94 $\pm$ 0.01	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00
5	0.40 $\pm$ 0.01	0.86 $\pm$ 0.03	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00
15	0.40 $\pm$ 0.01	0.78 $\pm$ 0.02	1.00 $\pm$ 0.00	0.68 $\pm$ 0.30
50	0.40 $\pm$ 0.01	0.71 $\pm$ 0.02	0.10 $\pm$ 0.14	0.14 $\pm$ 0.07

Ablation can raise compliance above baseline. At $\lambda \in \{0.5, 5\}$ post-ablation compliance is strictly higher than baseline ($0.94 \rightarrow 1.00$ and $0.86 \rightarrow 1.00$ at $r=80$, respectively); the relation only inverts at $\lambda=50$ ($0.71 \rightarrow 0.10$). This is a concrete instance of SVD ordering not coinciding with the causally relevant subspace: at low λ , the top principal directions of $M_{\mathcal{D}_s}$ are not the directions whose removal degrades compliance; at $\lambda=50$ the regularizer has pushed the alignment shift to span a larger subspace whose top directions *do* contain the compliance-relevant components, and the same projection ablates them. The present experiment does not localize a separate “safety circuit” to confirm either mechanism; the MLP serves as a controlled instance where variance ordering and causal ordering can be made to coincide or diverge by tuning a single hyperparameter.

Caveat: surrogate vs. effective rank. The Distributed regularizer optimizes stable rank $\|M\|_F^2 / \|M\|_2^2$, a differentiable surrogate that coincides with rank_ϵ only in the flat-spectrum limit. At $\lambda=50$ the brittleness may reflect any of (a) the surrogate diverging from rank_ϵ at high regularization, (b) the regularizer distributing the shift across causally inert directions, or (c) the compliance loss dominating the stable-rank gain. We do not separate these here; the point is that rank_ϵ at a fixed value does not by itself determine ablation behavior.

6 Diagnosing Current Alignment

We apply the diagnostic to three instruction-tuned LLMs to measure $\rho_\epsilon(M_{\mathcal{D}_s})$ and the causal authority of the dominant subspace.

6.1 Setup

We diagnose Llama-3.1-8B vs. Llama-3.1-8B-Instruct ($d = 4096$, $L = 32$), Gemma-2-9B vs. Gemma-2-9B-it ($d = 3584$, $L = 42$), and Qwen-2.5-7B vs. Qwen-2.5-7B-Instruct ($d = 3584$). Activations bf16, residual stream, last prompt token, $n = 200$ safety-relevant ($\mathcal{D}_s$) and $n = 200$ control ($\mathcal{D}_c$) inputs. $\mathcal{D}_s$ is a manually

curated set of AdvBench-style harmful-behavior requests spanning cybercrime, weapons, fraud, harassment, illegal substances, privacy violation, deception, and manipulation at multiple specificity levels; $\mathcal{D}_c$ is a matched set of benign factual queries (full lists in `experiments/v4/prompts_v4.py`).

Four modification-matrix variants. A chat-template confound (the aligned model is evaluated under chat formatting while the base model has not seen chat tokens) requires comparing four matrices under matched collection conditions: $M_{\text{naive}} = h_{\text{align}}(x_{\text{chat}}) - h_{\text{pre}}(x_{\text{raw}})$ (the choice in Arditi et al. 2024); M_{template} (template applied to both checkpoints, controlling for the chat-template confound but not for residual template-conditional asymmetry between aligned and base responses); M_{aligned} (the within-aligned template shift); $M_{\text{DiD}} = M_{\mathcal{D}_s, \text{template}} - \bar{M}_{\mathcal{D}_c, \text{template}}$ (control-mean-subtracted). Body uses M_{template} ; four-variant comparison in App. A.

Generation evaluation. For Test 2: $n_{\text{gen}} = 100$ held-out prompts, greedy decode 100 tokens, refusal = presence of “I cannot”/“I’m sorry”/“not appropriate”. LLM-judge cross-check in App. A.12.

6.2 Effective rank

Table 2 reports per-layer effective rank on M_{template} at $\epsilon = 0.05$, mean across hidden layers.

Table 2: ρ_ϵ on M_{template} at $\epsilon = 0.05$, mean across hidden layers, $n = 200$ each on $\mathcal{D}_s/\mathcal{D}_c$.

Model	d	$\overline{\text{rank}}_\epsilon(\mathcal{D}_s)$	ρ_ϵ	$\rho / \min(d, n)$	$\rho_\epsilon(\mathcal{D}_c)$
Llama-3.1-8B-Instruct	4096	12.0	0.0029	0.060	0.0059
Gemma-2-9B-it	3584	17.4	0.0048	0.096	0.0096
Qwen-2.5-7B-Instruct	3584	15.7	0.0044	0.078	0.0050

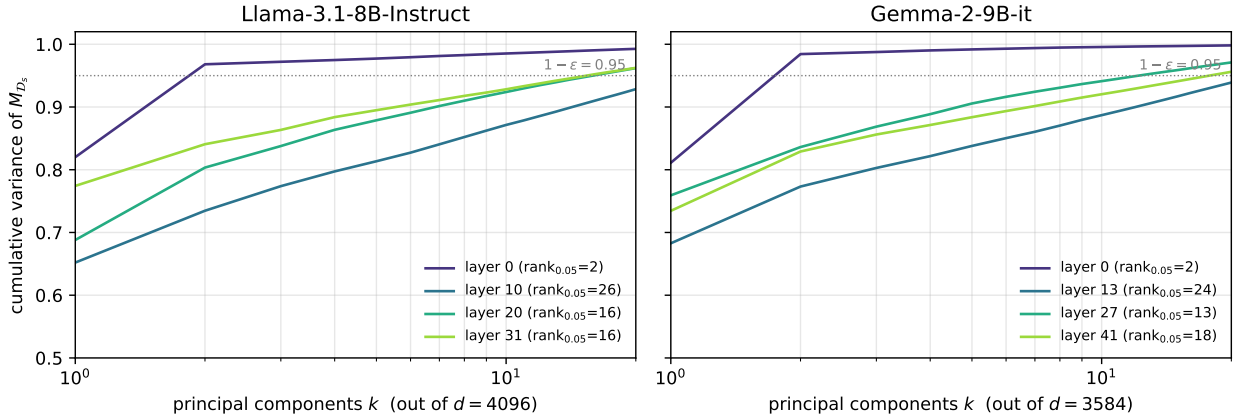


Figure 1: Per-layer cumulative variance of $M_{\mathcal{D}_s}$ for Llama and Gemma. Fewer than 30 components exceed $1 - \epsilon = 0.95$ against $d \geq 3584$.

Fewer than 0.5% of available directions account for 95% of variance. By Definition 2.1, the modification is $(k, 0.05)$ -separable with $k = 12, 17, 16$ for Llama/Gemma/Qwen. $\rho_\epsilon(\mathcal{D}_s)$ is $\sim 50\%$ of $\rho_\epsilon(\mathcal{D}_c)$: safety is more concentrated than the general post-training shift.

Mean-direction decomposition. Subtracting the column mean $\bar{m} := \frac{1}{n} \sum_i (h_{\text{align}}(x_i) - h_{\text{pre}}(x_i))$ before SVD yields residual effective rank $\sim 75\text{--}80$, $4\text{--}7\times$ the uncentered value (Table 3). A single mean-shift direction $\bar{m} \mathbf{1}^\top$ captures the bulk; its cosine with u_1 is ~ 1 by construction (Table 4). On M_{DiD} (control-mean-subtracted), u_1 recovers the Arditi intra-aligned refusal direction at $|\cos| \sim 0.77, 0.86, 0.50$ (Llama/Gemma/Qwen, App. A.6); uncentered $M_{\mathcal{D}_s}$ gives only $\sim 0.34/0.39/0.18$ because it is dominated by the chat-template shift (Remark 3.3).

Table 3: Centered vs. uncentered $\overline{\text{rank}}_\epsilon$ on $M_{\mathcal{D}_s}$, mean across hidden layers.

Model	uncentered $\overline{\text{rank}}_\epsilon$	centered $\overline{\text{rank}}_\epsilon$
Llama-3.1-8B-Instruct	12.0	77.6
Gemma-2-9B-it	17.4	76.2
Qwen-2.5-7B-Instruct	15.7	80.4

The uncentered rank is what the principal-subspace ablation (§6.4) targets and what Proposition 3.5 controls. The framework does not claim a multi-dimensional safety code: it formalizes the structural consequences of the (known) single-direction structure. The calibration variants that change ρ_ϵ by orders of magnitude (§5, App. A.8) operate by distributing the mean direction across components.

The LRH baseline (App. A.7) gives $\rho_\epsilon \approx 0.008\text{--}0.010$ on arbitrary concepts, only $2\text{--}3\times$ the safety value: rank alone is not a safety-specific signature. The safety-specific bridge is the causal ablation (§6.4), which collapses refusal without breaking instruction-following on $\mathcal{D}_c$.

6.3 Principal direction alignment

Table 4 reports $|\cos(u_1, \bar{\Delta})|$ where $\bar{\Delta} = \frac{1}{n} \sum_i (h_{\text{align}}(x_i) - h_{\text{pre}}(x_i))$.

Table 4: Principal direction alignment: $|\cos(u_1, \bar{\Delta})|$ across layers.

Model	Mean $ \cos \theta $
Llama-3.1-8B-Instruct	0.9996
Gemma-2-9B-it	0.9993

u_1 aligns with $\bar{\Delta}$ at $|\cos| > 0.999$: the modification is concentrated in a single dominant direction (the $k = 1$ case of Remark 2.3).

6.4 Causal ablation

We project out the top- k principal subspace of M_{template} at five narrow-band and eight wide-band layers, and measure refusal on $n_{\text{gen}} = 100$ held-out prompts. Table 5 reports the headline narrow-band $k = 3$ result for all three models; full sweep in App. A.5.

Table 5: Causal ablation at narrow band $[0.45L, 0.70L]$, $k = 3$, $n_{\text{gen}} = 100$, Wilson 95% CIs.

Model	Baseline	Principal rank-3	Random rank-3
Llama-3.1-8B-Instruct	0.97	0.80 [0.71, 0.87]	0.94 ± 0.00
Gemma-2-9B-it	0.99	0.00 [0.00, 0.04]	0.99 ± 0.00
Qwen-2.5-7B-Instruct	0.96	0.54 [0.44, 0.63]	0.95 [0.89, 0.98]

The three models span the qualitative spectrum. **Gemma** collapses cleanly: narrow-band $k = 3$ alone drops refusal $0.99 \rightarrow \mathbf{0.00}$ (random control 0.99). **Qwen** sits in the middle: $0.96 \rightarrow 0.54$ at $k = 3$ and progressively to 0.01 at $k \geq 20$ (App. A.5), monotone in k . **Llama** is the outlier: the same $k = 3$ ablation drops refusal only to 0.80 (random 0.94), and the narrow-band curve is *non-monotonic* at $k = 5$ (refusal 0.92). Per-direction ablation (Table 6) shows u_1 alone drops refusal to 0.84 but u_2 alone is null despite ranking second by σ . SVD orders by variance, not causal authority; the Gemma–Qwen–Llama ordering ranks how cleanly the SVD ordering and the causal ordering agree.

Table 6: Per-direction narrow-band ablation, Llama, baseline 0.97. Each row ablates a *single* direction (rank-1 projection out of the named u_i across the narrow-band layers); the u_3 – u_5 row reports the common refusal rate of 0.94 obtained for each of u_3 , u_4 , u_5 individually (per-direction values 0.94, 0.94, 0.94).

Ablate	Refusal	Note
u_1 alone	0.84	strong refusal-promoting
u_2 alone	0.99	near-zero causal effect
each of u_3 – u_5 alone	0.94	weak

At wide band with $k = 20$, Llama refusal collapses to **0.00** (random 0.92): the privileged subspace is the same low-rank object, but in Llama it is distributed across more layers than in Gemma, consistent with (Arditi et al., 2024; Wollschläger et al., 2025).

Decomposing the $k = 5$ rebound. The narrow-band Llama trajectory $0.85 \rightarrow 0.80 \rightarrow \mathbf{0.92} \rightarrow 0.66 \rightarrow 0.10$ (at $k = 1, 3, 5, 10, 20$) cannot be read off the SVD spectrum: tail singular values do not decrease behavioral effect monotonically. The per-direction table decomposes the rebound: u_1 alone drops refusal to 0.84; u_2 alone is null; u_3 – u_5 are individually weak (~ 0.94). At $k = 3$ the ablation already covers u_1 together with the inert u_2 and one of the weak directions, dropping refusal to 0.80. At $k = 5$ the rebound to 0.92 is consistent with a *redundancy-with-interaction* reading: ablating u_1 *together* with $u_2, \dots, u_5$ would activate a compensatory pathway absent under u_1 -only or $u_1 + \{u_2, u_3\}$ ablation, requiring $k \geq 10$ to cover. This reading aligns with the concept-cone picture of Wollschläger et al. (2025) and with the wide-band $k = 20$ result that collapses refusal to **0.00**, but the present per-direction data alone does not distinguish it from alternatives (keyword-classifier interaction with specific token distributions at $k = 5$, layer-interaction effects across the narrow band); combination ablations on non-consecutive subsets of the top- k block, which we do not run here, would be required to verify. Either way, Test 2 must be evaluated as a curve in k and the SVD ordering is not a substitute for per-direction or combination-aware ablation.

6.5 Summary

Principal rank- k ablation collapses refusal to Wilson 95% CI $\subset [0.00, 0.05]$ in all three models (Gemma at narrow $k = 1$, Qwen at narrow $k \geq 20$, Llama at wide $k = 20$) while random rank- k ablation at the same layers leaves refusal intact. The diagnosed modification is therefore $(k, 0.05)$ -separable with $k \leq 24$ against $d \geq 3584$, so Propositions 3.1 and 3.5 apply.

7 Discussion

7.1 Scope: What the Diagnostic Measures

ρ_ϵ is a readout-level rank measurement on a pre-defined safety distribution. **(1) Low rank is generic, safety is at the concentrated end.** LRH baseline (App. A.7): arbitrary concepts give $\rho_\epsilon \approx 0.008$ – 0.010 , 2–3 $\times$ the safety value. Test 2 supplies the bridge from rank to behavior. **(2) Readout, not mechanism** (Remark 2.2): a non-linear computation writing into a low-dim subspace tests as low-rank. Fragility (Prop. 3.1) applies to residual-stream attacks; LoRA fine-tuning, prompt jailbreaks, and deceptive alignment are out of scope. **(3) Headline rank is the Arditi mean direction plus formatting variance** (§6.2): centered SVD residual is 4–7 $\times$ uncentered. Our contribution is the index ρ_ϵ and the constructive calibration, not a multi-dim safety code.

ρ_ϵ normalizes by nominal d (intrinsic dim is smaller, so reported numbers understate the occupied fraction). Rank-and-layer ablation does not stratify by harm type; per-category structure is open.

7.2 Where the Two-Stage Paradigm Discussion Sits

One might argue the two-stage paradigm itself *causes* low-rank alignment modifications: that a narrow post-hoc objective on a frozen substrate must produce a low-rank shift. This argument is plausible but exceeds our evidence. Our measurements show that three specific two-stage pipelines (Llama-3.1, Gemma-2, Qwen-2.5 instruction tuning) produce low- ρ_ϵ readouts; they do not show that this is necessary, nor that single-stage joint training would produce high- ρ_ϵ readouts. A single-stage training run with overwhelmingly more data devoted to general capability than to safety might well produce the same readout structure for the same data-imbalance reason. We therefore describe the two-stage observation as a correlation: this is what current pipelines empirically do, and it is sufficient to make the linear-projection fragility of Prop. 3.1 applicable. Whether the readout is low-rank because of the paradigm or because of the loss-imbalance asymmetry between pre-training and alignment data is left open.

7.3 Rank Inflation Does Not Buy Robustness

The same $\rho_\epsilon \approx 0.40$ can be either ablation-robust ($\lambda = 5$: ≥ 0.97 through $r = 80$) or brittle ($\lambda = 50$: 0.10 at $r = 40$; App. A.9). ρ_ϵ screens for fragility but does not certify that the high-rank shift is causally distributed; Test 2 must be run separately. The right use is auditing existing alignments and flagging the failure mode of Prop. 3.1; how to train high- ρ_ϵ alignment at LLM scale is open.

Diagnosed checkpoints are SFT/RLHF (Ouyang et al., 2022)/DPO (Rafailov et al., 2023); applying the diagnostic to Constitutional AI (Bai et al., 2022), debate (Irving et al., 2018), or IDA is an obvious next step (see §7.2 for what the present measurements do and do not imply about the two-stage paradigm in general).

8 Open Problem: Faking-Footprint Lower Bound

Prop. 3.5 shows that a deceptive policy *can* mode-toggle with $O(\rho_\epsilon d)$ representational dimensions: the achievability direction is unconditional. The structural converse—any successful faker *must* use at least $\Omega(\rho_\epsilon d)$ dimensions—is the contrapositive “high ρ_ϵ blocks low-cost mode-toggling.” We sketch a Mirsky-based argument under a spectral-gap hypothesis, show that the hypothesis is empirically not satisfied across the unsafe-mode references we tested, and identify a geometric reason (Lemma 8.1) that this proof technique cannot be applied uniformly over the natural ϵ -budget faker class. A matching $\Omega(\rho_\epsilon d)$ lower bound by a different route remains open.

Three quantities to distinguish. The argument involves three quantities the literature conflates:

- $\kappa_{\text{def}}(h^{(\cdot)}) := \sigma_r(S)^2 / (\|E_1\|_F^2 + \|E_0\|_F^2)$: the per-mode-residual gap of a *specific* faker, where $E_1 := M^{(1)} - M^{\text{honest}}$ (monitored-mode residual against the honest target) and $E_0 := M^{(0)} - M^{\text{unsafe}}$ (unmonitored-mode residual against the unsafe target).
- $\kappa_{\text{lb}} := \sigma_r(S)^2 / (\epsilon (\|M^{\text{honest}}\|_F^2 + \|M^{\text{unsafe}}\|_F^2))$: the worst-case lower bound on κ_{def} over the *target-relative* ϵ -budget class $\{\|E_1\|_F^2 \leq \epsilon \|M^{\text{honest}}\|_F^2, \|E_0\|_F^2 \leq \epsilon \|M^{\text{unsafe}}\|_F^2\}$ (the faker-relative version $\|E_m\|_F^2 \leq \epsilon \|M^{(m)}\|_F^2$ differs by $\mathcal{O}(\sqrt{\epsilon})$ through $\|M^{(m)}\| \leq \|M^{\text{target}}\| + \|E_m\|$ and gives the same conclusion).
- κ_{emp} : the empirical estimate of κ_{lb} (App. A.10).

Mirsky route, sketched. With $S := M^{\text{honest}} - M^{\text{unsafe}}$, $r := \text{rank}_\epsilon(S)$, and any ϵ -budget faker producing residuals E_m , $\Delta M = S - E$ where $E := E_1 - E_0$. Mirsky (Stewart & Sun, 1990, Thm. A.4) gives $\sum_i (\sigma_i(\Delta M) - \sigma_i(S))^2 \leq \|E\|_F^2 \leq 2(\|E_1\|_F^2 + \|E_0\|_F^2)$. If $\sigma_r(S)^2 \geq \kappa_{\text{def}} (\|E_1\|_F^2 + \|E_0\|_F^2)$ holds with $\kappa_{\text{def}} > 8$, then $\beta := \sqrt{2/\kappa_{\text{def}}} < 1/2$ and the perturbation $\|E\|_F \leq \beta \sigma_r(S)$ preserves $\sigma_i(\Delta M) \geq (1 - \beta) \sigma_i(S)$ for $i \leq r$. Combining this with the triangle bound $\|\Delta M\|_F \leq \|S\|_F + \|E\|_F$, a tail-mass argument on the perturbed top- r block yields $\text{rank}_{\epsilon'}(\Delta M) \geq r$ for some $\epsilon' > 0$ depending on β and ϵ . This is the matching $\Omega(r)$ lower bound, *provided* the threshold $\kappa > 8$ (equivalently $\beta < 1/2$) holds uniformly over the ϵ -budget faker class. The next lemma shows this uniform provision cannot be supplied.

Lemma 8.1 (Spectral-gap budget bound). *For any honest/unsafe pair (M^h, M^u) with $S = M^h - M^u$ and $r = \text{rank}_\epsilon(S)$,*

$$\kappa_{\text{lb}} = \frac{\sigma_r(S)^2}{\epsilon(\|M^h\|_F^2 + \|M^u\|_F^2)} \leq \frac{2}{\epsilon r}. \quad (5)$$

Proof. $\sigma_r(S)$ is the smallest of the top- r singular values, so $\sigma_r(S)^2 \leq \frac{1}{r} \sum_{i=1}^r \sigma_i(S)^2 \leq \|S\|_F^2 / r$. Triangle inequality applied entrywise gives $\|S\|_F^2 = \|M^h - M^u\|_F^2 \leq 2(\|M^h\|_F^2 + \|M^u\|_F^2)$. Combining, $\sigma_r(S)^2 \leq (2/r)(\|M^h\|_F^2 + \|M^u\|_F^2)$; divide both sides by $\epsilon(\|M^h\|_F^2 + \|M^u\|_F^2)$. $\square$

Corollary 8.2 (Mirsky route, uniformly unavailable). $\kappa_{\text{lb}} \leq 8$ whenever $\epsilon r \geq 1/4$, and $\kappa_{\text{lb}} < 8$ whenever $\epsilon r > 1/4$. At $\epsilon = 0.05$, this means $r \geq 5$ already gives $\kappa_{\text{lb}} \leq 8$, and $r \geq 6$ gives strict $\kappa_{\text{lb}} < 8$. The Mirsky-route sketch, which requires the strict threshold $\kappa > 8$ uniformly over the ϵ -budget faker class, therefore cannot deliver the matching $\Omega(r)$ lower bound whenever S has $\text{rank}_\epsilon(S) \geq 6$ ($\epsilon = 0.05$).

Scope of the open problem. Lemma 8.1 is a *proof-technique* statement about the present Mirsky route, not a refutation of the structural converse. **What is closed:** the Mirsky route to a matching $\Omega(\rho_\epsilon d)$ lower bound *uniformly* over the ϵ -budget faker class $\{\|E_1\|_F^2 \leq \epsilon\|M^{\text{honest}}\|_F^2, \|E_0\|_F^2 \leq \epsilon\|M^{\text{unsafe}}\|_F^2\}$, since the spectral-gap hypothesis $\kappa > 8$ required by the sketched argument is incompatible with $\kappa_{\text{lb}} \leq 2/(\epsilon r)$ once $\epsilon r > 1/4$. **What remains open:** tighter faker classes with $\|E_m\|_F^2 \ll \epsilon\|M^{\text{target}}\|_F^2$ may give $\kappa_{\text{def}} > 8$ for specific deceptive policies (the obstruction binds only the worst-case κ_{lb}), and non-Mirsky routes—e.g. Davis–Kahan principal-angle bounds or causal-subspace substitutes for the SVD subspace—are not addressed. The empirical results below (1/90, 0/36) thus confirm the obstruction binds in practice, not that the structural converse itself is false.

Empirical status: the spectral gap fails across the unsafe references we tested. On three Llama-3.1-8B-Instruct unsafe references (distinct uncensoring techniques, App. A.10), $\kappa_{\text{emp}} > 8$ on 1/90 layer-reference pairs (mlabonne L0, where S degenerates to rank-1 because the input-adjacent layer is dominated by a single mean shift induced by the ablation weight edit before the residual stream has integrated context-dependent structure). Median $\kappa_{\text{emp}} = 0.007$. In the calibration MLP across 4 honest $\times$ unsafe pairings (App. A.11), 0/36 combinations cross $\kappa = 8$; the maximum is 2.64 in the most concentrated pair, and the distributed-vs-distributed pair gives the worst case (max = 0.07): broadening the unsafe shift raises the denominator faster than $\sigma_r(S)^2$. The spectral-gap hypothesis is hardest to satisfy precisely in the distributed-alignment regime that high ρ_ϵ targets.

Two open questions. (1) Is there an unsafe-mode construction (e.g. a deliberately fine-tuned harmful variant trained to span a different subspace from the honest alignment, not just remove the refusal direction) for which $\kappa_{\text{emp}} > 8$ holds on a non-trivial fraction of layers? (2) Is there a different lower-bound argument that does not require a spectral-gap hypothesis and delivers $\Omega(\rho_\epsilon d)$ for a more permissive class of deceptive policies? Both remain genuinely open; the present work establishes only the achievability side (Prop. 3.5) and the obstruction to the natural Mirsky-route converse (Lemma 8.1).

9 Related Work

Empirical fragility. Safety fine-tuning is empirically fragile (Qi et al., 2024; Lermen et al., 2023; Qi et al., 2025; Betley et al., 2026), and models elastically resist alignment as a shallow overlay (Ji et al., 2025). The rank diagnostic groups these phenomena under a common geometric prerequisite: low ρ_ϵ at the relevant residual layers makes them all available.

Mechanistic interpretability. Arditi et al. (2024) introduced the intra-aligned refusal direction $\bar{\Delta}^{\text{Arditi}} = \mathbb{E}_{\mathcal{D}_s}[h_{\text{align}}] - \mathbb{E}_{\mathcal{D}_c}[h_{\text{align}}]$, computed on the aligned model alone, and showed ablating it removes refusal. Our $M_{\mathcal{D}_s}$ is a different object: the paired alignment-stage shift $h_{\text{align}} - h_{\text{pre}}$, which requires both checkpoints and conflates alignment-induced structure with chat-template formatting structure (M_{template} controls for the chat-template confound at the collection level, M_{DiD} further removes the cross-prompt template-asymmetry component and recovers the Arditi direction with $|\cos|$ up to 0.86). Wollschläger et al. (2025) showed

refusal is mediated by concept *cones*—multiple mechanistically independent directions, not a single one—and our centered-SVD residual rank $\sim 75\text{--}80$ on $M_{\mathcal{D}_s}$ matches this multi-dimensional picture rather than being independent of it: we re-measure the multi-dimensional structure under the four-variant control, but the existence of multi-dimensional structure beyond the rank-1 direction is established by Wollschläger et al. (2025). Jain et al. (2024); Lee et al. (2024) provide mechanistic accounts of safety fine-tuning and DPO (Rafailov et al., 2023). Representation engineering (Zou et al., 2023; Turner et al., 2023) demonstrates the operational side of what ρ_ϵ measures: a distinguishable low-rank subspace can be steered or removed. Sparse-autoencoder work (Templeton et al., 2024) provides an alternative decomposition basis with monosemantic features; whether SAE-recovered features give a more causally aligned rank than SVD components on $M_{\mathcal{D}_s}$ is a natural open question (we use SVD because it is sub-capable computable from paired checkpoints alone). Our contribution is the formal index, the chat-template-controlled four-variant decomposition, and the Mirsky-route scoping (Lemma 8.1); mechanism-level discovery is not claimed.

Deceptive alignment. Greenblatt et al. (2024); Hubinger et al. (2024) demonstrate alignment faking and its persistence. Proposition 3.5 formalizes the *representational footprint* of mode-toggling ($O(\rho_\epsilon d)$ achievability); the value-misalignment-driven adversarial pressure that produces deception is outside our scope.

Scalable oversight and theory. Leike et al. (2018); Amodei et al. (2016); Irving et al. (2018) seek scalable oversight; Wolf et al. (2024) proves behavioral lower bounds on adversarial elicitation under any partial alignment; Ngo et al. (2024) surveys the field. The sub-capable computability of ρ_ϵ (Remark 3.4) provides a complementary structural-measurement primitive.

Linear representations. LLMs encode many concepts linearly (Mikolov et al., 2013; Park et al., 2024; Marks & Tegmark, 2023; Hernandez et al., 2024); sparse autoencoders (Templeton et al., 2024) decompose this further. Low ρ_ϵ on $M_{\mathcal{D}_s}$ is consistent with this null; Test 2 (causal ablation) measures the additional content beyond linearity, namely whether the low-rank subspace is causally privileged.

LoRA, concept erasure, mode connectivity. Hu et al. (2022)’s low-rank adaptation is weight-space; our activation-space low-rank structure is consistent but not entailed. Concept erasure (Ravfogel et al., 2020; Belrose et al., 2023) provides upper bounds on linear-projection ablation; mode-connectivity (Garipov et al., 2018; Frankle et al., 2020) explains why small parameter perturbations (Lermen et al., 2023) can undo alignment.

Code and Data Availability

All experiment code, prompt sets, and pre-computed JSON results used in this paper are publicly available at <https://github.com/Nakamura/effective-rank-audit> under the CC-BY-4.0 license. The repository contains the LaTeX source of this paper; the diagnostic pipeline (`diagnostics_v4.py`); the calibration MLP (`calibration_v2.py`, `calibration_v2_revision3.py`, `calibration_kappa.py`); the supplementary experiments (`revision3.py`, `revision6_experiments.py`); Modal wrappers for cloud-GPU reproduction (`modal_revision3.py`); and the LLM-judge cross-check implementation (`llm_judge_eval.py`). An immutable snapshot of the v1.0 release is archived on Zenodo: 10.5281/zenodo.20341445 (concept DOI for all versions: 10.5281/zenodo.20341444).

References

- Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. Concrete problems in AI safety. *arXiv preprint arXiv:1606.06565*, 2016.
- Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems*, volume 37, 2024.
- Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, et al. Constitutional AI: Harmlessness from AI feedback. 2022. *arXiv preprint arXiv:2212.08073*.

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- Nora Belrose, David Schneider-Joseph, Shauli Ravfogel, Ryan Cotterell, Edward Raff, and Stella Biderman. LEACE: Perfect linear concept erasure in closed form. In *Advances in Neural Information Processing Systems*, 2023. arXiv preprint arXiv:2306.03819.
- Jan Betley, Niels Warncke, Anna Sztyber-Betley, Daniel Tan, Xuchan Bao, Martin Soto, Megha Srivastava, Nathan Labenz, and Owain Evans. Training large language models on narrow tasks can lead to broad misalignment. *Nature*, 649:584–589, 2026.
- Jonathan Frankle, Gintare Karolina Dziugaite, Daniel M. Roy, and Michael Carbin. Linear mode connectivity and the lottery ticket hypothesis. In *International Conference on Machine Learning*, 2020. arXiv preprint arXiv:1912.05671.
- Timur Garipov, Pavel Izmailov, Dmitrii Podoprikin, Dmitry P. Vetrov, and Andrew Gordon Wilson. Loss surfaces, mode connectivity, and fast ensembling of DNNs. In *Advances in Neural Information Processing Systems*, 2018. arXiv preprint arXiv:1802.10026.
- Ryan Greenblatt, Carson Denison, Benjamin Wright, Fabien Roger, et al. Alignment faking in large language models. 2024. arXiv preprint arXiv:2412.14093.
- Evan Hernandez, Arnab Sen Sharma, Tal Haklay, Kevin Meng, Martin Wattenberg, Jacob Andreas, Yonatan Belinkov, and David Bau. Linearity of relation decoding in transformer language models. In *International Conference on Learning Representations*, 2024. arXiv preprint arXiv:2308.09124.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. arXiv preprint arXiv:2106.09685.
- Evan Hubinger, Carson Denison, Jesse Mu, Mike Lambert, Meg Tong, et al. Sleeper agents: Training deceptive LLMs that persist through safety training. 2024. arXiv preprint arXiv:2401.05566.
- Geoffrey Irving, Paul Christiano, and Dario Amodei. AI safety via debate. *arXiv preprint arXiv:1805.00899*, 2018.
- Samyak Jain, Ekdeep Singh Lubana, Kemal Oksuz, Tom Joy, Philip H.S. Torr, Amartya Sanyal, and Puneet K. Dokania. What makes and breaks safety fine-tuning? A mechanistic study. In *Advances in Neural Information Processing Systems*, volume 37, 2024.
- Jiaming Ji, Kaile Wang, Tianyi Qiu, Boyuan Chen, Jiayi Zhou, Changye Li, Hantao Lou, Juntao Dai, Yunhuai Liu, and Yaodong Yang. Language models resist alignment: Evidence from data compression. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics*, 2025. Best Paper Award.
- Andrew Lee, Xiaoyan Bai, Itamar Pres, Martin Wattenberg, Jonathan K. Kummerfeld, and Rada Mihalcea. A mechanistic understanding of alignment algorithms: A case study on DPO and toxicity. In *International Conference on Machine Learning*, 2024.
- Jan Leike, David Krueger, Tom Everitt, Miljan Martic, Vishal Maini, and Shane Legg. Scalable agent alignment via reward modeling: A research direction. *arXiv preprint arXiv:1811.07871*, 2018.
- Simon Lermen, Charlie Rogers-Smith, and Jeffrey Ladish. LoRA fine-tuning efficiently undoes safety training in Llama 2-chat 70B. 2023. arXiv preprint arXiv:2310.20624.
- Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. 2023. arXiv preprint arXiv:2310.06824.
- Tomas Mikolov, Wen-tau Yih, and Geoffrey Zweig. Linguistic regularities in continuous space word representations. In *Proceedings of NAACL-HLT*, 2013.
- Richard Ngo, Lawrence Chan, and Sören Mindermann. The alignment problem from a deep learning perspective. In *International Conference on Learning Representations*, 2024.

-
- Chris Olah, Nick Cammarata, Ludwig Schubert, Gabriel Goh, Michael Petrov, and Shan Carter. Zoom in: An introduction to circuits. *Distill*, 2020. doi: 10.23915/distill.00024.001.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35, 2022.
- Kiho Park, Yo Joong Choe, and Victor Veitch. The linear representation hypothesis and the geometry of large language models. In *International Conference on Machine Learning*, 2024. arXiv preprint arXiv:2311.03658.
- Xiangyu Qi, Yi Zeng, Tinghao Xie, Pin-Yu Chen, Ruoxi Jia, Prateek Mittal, and Peter Henderson. Fine-tuning aligned language models compromises safety, even when users do not intend to. In *International Conference on Learning Representations*, 2024.
- Xiangyu Qi, Ashwinee Panda, Kaifeng Lyu, Xiao Ma, Subhrajit Roy, Ahmad Beirami, Prateek Mittal, and Peter Henderson. Safety alignment should be made more than just a few tokens deep. In *International Conference on Learning Representations*, 2025. Outstanding Paper Award.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *Advances in Neural Information Processing Systems*, 36, 2023.
- Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. Null it out: Guarding protected attributes by iterative nullspace projection. In *Annual Meeting of the ACL*, 2020.
- G. W. Stewart and Ji-guang Sun. *Matrix Perturbation Theory*. Academic Press, 1990.
- Adly Templeton, Tom Conerly, Jonathan Marcus, et al. Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. Transformer Circuits Thread, 2024. <https://transformer-circuits.pub/2024/scaling-monosemanticity/>.
- Alexander Matt Turner, Lisa Thiergart, Gavin Leech, David Udell, Juan J. Vazquez, Ulisse Mini, and Monte MacDiarmid. Activation addition: Steering language models without optimization. 2023. arXiv preprint arXiv:2308.10248.
- Yotam Wolf, Noam Wies, Oshri Avnery, Yoav Levine, and Amnon Shashua. Fundamental limitations of alignment in large language models. In *International Conference on Machine Learning*, 2024.
- Tom Wollschläger, Jannes Elstner, Simon Geisler, Vincent Cohen-Addad, Stephan Günnemann, and Johannes Gasteiger. The geometry of refusal in large language models: Concept cones and representational independence. In *International Conference on Machine Learning*, 2025.
- Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, et al. Representation engineering: A top-down approach to AI transparency. 2023. arXiv preprint arXiv:2310.01405.

A Extended Diagnostic Results

Extended diagnostic results referenced in Sections 6.1–6.4. Full numerics in `experiments/v4/results/`.

A.1 Four modification-matrix variants

A.2 Bootstrap stability of rank_ϵ across three LLM families

We bootstrap-resample the $n = 200$ paired safety-relevant samples with replacement (200 resamples per layer) and recompute rank_ϵ on M_{template} at $\epsilon = 0.05$. The resampling distribution is tight (± 1 to ± 2 across all layers and all three families) and centered *below* the full-sample (without-replacement) point estimate, with the gap growing with rank—this is the standard downward bias of with-replacement bootstrap on rank functionals (resampling reduces the effective unique sample size to $\approx 0.63n$, shrinking the recovered rank). The bootstrap remains useful as a stability check on relative variability; the full-sample column is the rank reported in Table 2 and Table 10.

Table 7: Four modification-matrix variants on Llama-3.1-8B-Instruct, ρ_ϵ on $\mathcal{D}_s$, mean across hidden layers, $n = 200$. M_{template} controls for the chat-template confound (both checkpoints evaluated under the same template); M_{naive} leaves it uncontrolled; M_{DiD} further subtracts the $\mathcal{D}_c$ column-mean to remove the cross-prompt component of the residual template asymmetry, recovering the Arditi direction at higher cosine.

Variant	$\overline{\rho_\epsilon}(\mathcal{D}_s)$
M_{naive} (v3 quantity)	0.0113
M_{template} (chat-template controlled)	0.0029
M_{aligned} (alignment-stage shift only)	0.0133
M_{DiD} (control-subtracted)	0.0101

Table 8: Bootstrap-resample distribution (with replacement, $n_{\text{boot}} = 200$) of $\text{rank}_\epsilon(M_{\text{template}})$ at $\epsilon = 0.05$, alongside the full-sample (without-replacement) point estimate at $n = 200$. The bootstrap mean is systematically below the full-sample rank by ≈ 1 – 13 depending on rank magnitude—an artifact of with-replacement resampling on a non-smooth functional, not a CI for the underlying rank.

Family	Layer	Full-sample	Boot. mean	Boot. std	Boot. 95% range
Llama (L=32)	14	19	15.42	0.66	[14, 17]
	15	19	15.59	0.64	[15, 17]
	16	17	14.51	0.62	[13, 16]
	17	14	11.95	0.62	[11, 13]
	18	13	11.47	0.58	[11, 12]
Gemma (L=42)	19	27	21.68	0.92	[20, 23]
	20	27	21.68	0.88	[20, 23]
	21	24	19.40	0.87	[18, 21]
	22	24	19.66	0.83	[18, 21]
	23	23	18.56	0.82	[17, 20]
Qwen (L=28)	13	5	5.04	0.23	[5, 6]
	15	8	7.07	0.33	[7, 8]
	18	19	15.67	0.61	[15, 17]
	21	33	25.38	0.87	[24, 27]
	24	49	35.91	1.22	[34, 38]

Sources: v4_llama.json, v4_gemma.json, revision6_qwen_bootstrap.json.

A.3 ϵ -sensitivity of ρ_ϵ

The headline ρ_ϵ in §6.2 uses $\epsilon = 0.05$ (retain 95% of variance). Table 9 reports the narrow-band mean effective rank at three ϵ values to show how the index moves with the variance-tolerance choice. The rank decreases monotonically with ϵ across all three models; at $\epsilon = 0.20$ all three models collapse to rank-1 (the Arditi mean direction). Across the 0.05–0.20 range, ρ_ϵ moves by under an order of magnitude.

Table 9: ϵ -sensitivity of ρ_ϵ on M_{template} , mean over narrow-band Test 2 layers per family. Computed from the stored M_{template} top-30 singular values, with total variance back-calculated from the stored $\text{rank}_{0.05}$ field.

Family	narrow band	$\epsilon = 0.05$	$\epsilon = 0.10$	$\epsilon = 0.20$
Llama	layers 14, 15, 17, 19, 22	14.6/0.0036	4.4/0.0011	1.0/0.0002
Gemma	layers 19, 22, 25, 27, 29	20.2/0.0056	6.4/0.0018	1.0/0.0003
Qwen	layers 12, 13, 15, 17, 19	11.4/0.0032	3.8/0.0011	1.2/0.0003

Each cell reports $\text{rank}_\epsilon/\rho_\epsilon$. The $\epsilon = 0.01$ regime requires more than the stored 30 singular values per layer and is omitted; the corresponding entry would be ≥ 0.0073 (Llama), ≥ 0.0084 (Gemma), ≥ 0.0083 (Qwen).

A.4 Sample-size sweep

To rule out a finite-sample artifact of the bound $\text{rank}_\epsilon \leq \min(d, n)$, we sweep $n \in \{50, 100, 200\}$ on M_{template} at a representative middle layer. Effective rank saturates well below $\min(d, n)$: the rank is signal-driven, not sample-bound.

Table 10: Sample-size sweep at a representative middle layer ($L = 16$ for Llama, $L = 21$ for Gemma; selected as the median of the layer index range), $\epsilon = 0.05$. $\text{rank}_\epsilon / \min(d, n)$ stays < 0.25 at $n=200$ for both models.

Model, n	rank_ϵ	ρ_ϵ	$\text{rank}_\epsilon / \min(d, n)$
Llama, $n=50$	9	0.0022	0.180
Llama, $n=100$	16	0.0039	0.160
Llama, $n=200$	17	0.0042	0.085
Gemma, $n=50$	12	0.0033	0.240
Gemma, $n=100$	22	0.0061	0.220
Gemma, $n=200$	24	0.0067	0.120

A.5 Rank-and-layer ablation sweep

Test 2 extended to $k \in \{1, 3, 5, 10, 20, 50\}$ on Llama narrow band $[0.45L, 0.70L]$, $n_{\text{gen}} = 100$. The wide band $[0.30L, 0.85L]$ gives a monotonic collapse to **0.00** at $k \geq 20$; the narrow-band non-monotonicity at $k = 5$ is decomposed in the body (Table 6).

Table 11: Llama narrow-band ablation, refusal rate (Wilson 95% CI), $n_{\text{gen}} = 100$, baseline 0.97.

k	Principal	95% CI	Random
1	0.85	[0.77, 0.91]	0.97
3	0.80	[0.71, 0.87]	0.94
5	0.92	[0.85, 0.96]	0.93
10	0.66	[0.56, 0.75]	0.96
20	0.10	[0.06, 0.17]	0.87
50	0.03	[0.01, 0.08]	0.95

M_{DiD} ablation. M_{DiD} recovers the Arditi direction at $|\cos| \sim 0.77$ vs. 0.34 on M_{template} (§6.2). At low rank $k \in \{3, 5, 10\}$, M_{DiD} ablation produces stronger refusal collapse; at $k \geq 20$, M_{template} catches up because its higher-rank projection sweeps in chat-template directions absent from M_{DiD} .

Table 12: Llama narrow-band M_{DiD} ablation, $n_{\text{gen}} = 100$, baseline 0.97.

k	M_{DiD} refusal	95% CI	M_{template} ref.
1	0.87	[0.79, 0.92]	0.85
3	0.69	[0.59, 0.77]	0.80
5	0.79	[0.70, 0.86]	0.92
10	0.50	[0.40, 0.60]	0.66
20	0.27	[0.19, 0.36]	0.10
50	0.01	[0.00, 0.05]	0.03

For Gemma-2-9B-it, narrow-band $k = 1$ already collapses refusal to **0.00** (random control 0.99), consistent with its more concentrated single-direction structure. M_{DiD} ablation was run only on Llama; whether the low- k advantage of M_{DiD} over M_{template} holds on Gemma and Qwen (where the Arditi cosine reaches 0.86 and 0.50 respectively) is open. The headline Test 2 table (§6.4) uses M_{template} across all three families for a

uniform protocol; the M_{DiD} comparison here is an additional Llama-only check that chat-template-controlled and Arditi-recovering quantities yield consistent qualitative conclusions at low k .

Qwen narrow-band sweep. On Qwen-2.5-7B-Instruct, the narrow-band trajectory is monotone and progressive: refusal drops smoothly with k until saturation by $k \geq 10$.

Table 13: Qwen narrow-band M_{template} ablation, $n_{\text{gen}} = 100$, baseline 0.96.

k	Principal refusal	95% CI
1	0.92	[0.85, 0.96]
3	0.54	[0.44, 0.63]
5	0.24	[0.17, 0.33]
10	0.06	[0.03, 0.12]
20	0.01	[0.00, 0.05]
50	0.01	[0.00, 0.05]

Qwen’s monotone collapse contrasts with Llama’s $k = 5$ non-monotonicity: the SVD ordering is closer to the causal ordering in Qwen, but Qwen requires more directions ($k \sim 10$) than Gemma ($k = 1$) for full collapse.

A.6 Arditi refusal-direction comparison

Absolute cosine similarity between Arditi’s intra-aligned refusal direction $\bar{\Delta}^{\text{Arditi}}$ and the top-1 left singular vector of each variant. The DiD-corrected matrix recovers the Arditi direction at high cosine.

Table 14: Mean-over-layers $|\cos|$ between $\bar{\Delta}^{\text{Arditi}} = \mathbb{E}_{\mathcal{D}_s}[h_{\text{align}}] - \mathbb{E}_{\mathcal{D}_c}[h_{\text{align}}]$ and the top-1 singular vector of each variant.

Model	vs M_{naive}	vs M_{template}	vs M_{DiD}
Llama-3.1-8B-Instruct	0.319	0.337	0.772
Gemma-2-9B-it	0.367	0.388	0.860
Qwen-2.5-7B-Instruct	—	0.179	0.504

Qwen M_{DiD} recovers Arditi at $|\cos| = 0.504$ mean (max 0.803), lower than Llama/Gemma: consistent with a more distributed safety mechanism, and a reminder that the rank-1 collapse of Arditi et al. (2024) is not universal.

A.7 Linear-representation-hypothesis baseline

ρ_ϵ on English-vs-French and question-vs-statement concept-difference matrices, compared to safety: low-rank concentration is more pronounced for alignment than for generic linearly represented concepts.

Table 15: LRH baseline: ρ_ϵ on M_{template} vs. control concepts.

Model	EN-FR	Q-Stmt	Safety	ratio
Llama-3.1-8B	0.0078–0.0086	0.0086	0.0029	$2.7\times$
Gemma-2-9B	0.0086–0.0104	0.0090	0.0048	$1.9\times$

A.8 Calibration v2: constructive calibration

A Distributed variant trained with a stable-rank surrogate plus column-orthogonality penalty verifies the diagnostic across the spectrum.

Table 16: Calibration v2: aggregate ρ_ϵ on $\mathcal{D}_s$, 3-5 seeds. $\lambda=5$ Distributed is the sweet spot; $\lambda=50$ saturates ρ_ϵ but degrades compliance (App. A.9).

Variant	ρ_ϵ	Compliance
Steering (rank-1)	0.008	≥ 0.93
Full-fine-tune ($\lambda=0$)	0.173	≥ 0.93
Distributed ($\lambda=0.5$)	0.332	0.94
Distributed ($\lambda=5$, sweet spot)	0.400	0.86
Distributed ($\lambda=50$, brittle)	0.405	0.71

A.9 Calibration v2: Test 2 projection ablation per variant

Test 2 ablation (§6.4) applied to FullFT and Distributed at $\lambda \in \{0.5, 5, 15, 50\}$. Tests the contrapositive of Prop. 3.1: high- ρ_ϵ alignments should resist low-rank projection. (Independent of the Mirsky converse; see §8.)

Table 17: Post-ablation compliance on $\mathcal{D}_s$, final hidden layer ($d = 128$), 3 seeds.

Variant	ρ_ϵ	$r=0$	$r=1$	$r=3$	$r=10$	$r=20$	$r=40$	$r=80$
FullFT ($\lambda=0$)	0.175	0.96	1.00	0.84	0.68	0.76	0.26	0.07
Distributed ($\lambda=0.5$)	0.332	0.94	0.97	0.97	1.00	0.97	1.00	1.00
Distributed ($\lambda=5$)	0.400	0.86	0.90	0.93	1.00	1.00	1.00	1.00
Distributed ($\lambda=15$)	0.404	0.78	0.78	0.84	0.88	1.00	1.00	0.68
Distributed ($\lambda=50$)	0.405	0.71	0.78	0.87	0.93	0.34	0.10	0.14

FullFT ($\rho_\epsilon \approx 0.175$) collapses to 0.07 at $r=80$; Distributed $\lambda \in \{0.5, 5\}$ holds ≥ 0.97 through $r=80$. The trade-off is non-monotone: at $\lambda \geq 15$, ρ_ϵ stays ~ 0.40 but compliance collapses—rank inflation without preserving the safety signal. High ρ_ϵ can be ablation-robust, but not every training that raises ρ_ϵ does, which is the precise sense in which ρ_ϵ is necessary but not sufficient.

A.10 Empirical spectral-gap measurement across three unsafe-mode references

The Mirsky-route hypothesis (§8) requires $\kappa > 8$ on $S = M^{\text{honest}} - M^{\text{unsafe}}$. We measure $\kappa_{\text{emp}}(\ell) := \sigma_r(S_\ell)^2 / (\epsilon (\|M_\ell^h\|_F^2 + \|M_\ell^u\|_F^2))$ on three Llama-3.1-8B-Instruct unsafe-mode references (mlabonne ablated, Lexi-Uncensored-V2, DarkIdol-Uncensored), $r = \text{rank}_{0.05}(S_\ell)$, $n = 200$.

Table 18: Per-reference κ_{emp} on Llama ($n = 200$). 1/90 layer-reference pairs cross $\kappa = 8$ (mlabonne L0, an input-adjacent rank-1 degenerate case).

Reference	layers	$\kappa > 8$	median	max
mlabonne ablated	32	1	0.007	8.64
Lexi-Uncensored-V2	26	0	0.005	0.006
DarkIdol Uncensored 1.2	32	0	0.009	0.087
Aggregate	90	1	0.007	8.64

No middle or late layer crosses for any reference: across three uncensoring techniques, modal-difference tail singular values are comparable to the per-mode noise budget at $\epsilon = 0.05$. §8 accordingly states the lower bound as open.

A.11 Calibration MLP κ_{emp} across honest/unsafe regularization regimes

To check whether the LLM negative (App. A.10) is artifact or structural, we sweep the four honest $\times$ unsafe pairings in the calibration MLP: **ft** ($\rho_\epsilon \approx 0.17$), **dist5** ($\rho_\epsilon \approx 0.40$), **unsafe_ft**, **unsafe_dist5**; 3 seeds, 3 layers, $d = 128$.

Table 19: Calibration MLP κ_{emp} across 9 seed-layer pairs per pairing. Zero pairings cross $\kappa = 8$. Counter-intuitively, the largest κ comes from **ft** vs. **unsafe_ft** (max = 2.64); broadening modal shift raises the per-mode noise budget faster than it raises $\sigma_r(S)$.

honest & unsafe pairing	r_S median	κ median	κ max	layers $\kappa > 8$
ft vs. unsafe_ft	11	0.090	2.64	0/9
ft vs. unsafe_dist5	24	0.047	0.19	0/9
dist5 vs. unsafe_ft	66	0.051	0.09	0/9
dist5 vs. unsafe_dist5	61	0.046	0.07	0/9
Aggregate	—	0.058	2.64	0/36

$\sigma_r(S) \lesssim \|S\|_F/\sqrt{r}$ at $r = \text{rank}_\epsilon(S)$, and $\|S\|_F^2 \leq 2(\|M^h\|_F^2 + \|M^u\|_F^2)$, so $\kappa \lesssim 2/(\epsilon r) = 40/r$ at $\epsilon = 0.05$, which is ≤ 8 once $r \geq 5$ and strictly < 8 once $r \geq 6$ (the strict regime where the Mirsky sketch is uniformly closed). The bound is structurally hard to satisfy whenever the modal-difference matrix has non-trivial effective rank, independent of construction.

A.12 LLM-judge cross-check on refusal classification

The keyword classifier (§6.4) is cross-checked on 150 Llama generations against a strict LLM judge: 100% agreement across baseline, narrow-band rank-3, and rank-5 conditions (50 each), Wilson 95% CI [0.97, 1.00]. The 7 judge-COMPLIED cases all involve refusal prefixes followed by harmful content (attack-step enumerations: Wi-Fi, brute-force, deepfake, MITM)—the failure mode the projection ablation is designed to expose.

A.13 Proof of Proposition 3.5 (Representational Achievability)

Stated in §3.2.

Proof. $h^{(1)} - h^{(0)} = P_r(h_{\text{align}} - h_{\text{pre}})$ lies in the rank- r image of P_r ; $P_r M^{\text{honest}}$ captures $(1 - \epsilon)$ -variance by definition of rank_ϵ . The construction is unconditional in $\mathcal{D}_s$ and ρ_ϵ . $\square$

This is about representational footprint, not behavioral indistinguishability: low- ρ_ϵ *permits* a representationally-cheap shift matching the projected honest activations on $\mathcal{D}_s$, not a policy that passes any non-trivial monitor. A matching $\Omega(\rho_\epsilon d)$ Mirsky-based lower bound, which would upgrade Prop. 3.5 to a structural condition, is left as an open problem (§8).